Vinayak Kamat

ML Engineer | Applied Machine Learning and Financial Engineering

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Summary

Machine learning engineer with 1 year 8 months at KAN Innovations and an MSc in Financial Engineering in progress. Experience building and evaluating PyTorch models, improving model calibration, and putting ML into production with Python. Qualified GATE. Seeking to apply machine learning and statistical thinking to financial markets.

Education

MSc **WorldQuant University**, Financial Engineering — Remote, Washington DC — Sept 2025 – May 2027

B.Tech **Punjab Technical University**, Computer Science — Punjab, India — Sept 2021 – May 2025

Skills

Languages: Python, C++, JavaScript, Dart

Machine Learning: PyTorch, Transformers, model calibration, model evaluation, ONNX, spaCy, OpenCV

Data: Pandas, NumPy, MLflow (experiment tracking and model versioning)

Engineering: FastAPI, Flask, Docker, AWS (Lambda, S3, ECR, DynamoDB)

Experience

KAN Innovations, Junior AI/ML Engineer — Mumbai, Maharashtra — May 2025 – present — 1 year 6 months

- Improved model calibration so prediction confidence better reflects real accuracy, making outputs more reliable in production
- Built and evaluated Retrieval-Augmented Generation (RAG) pipelines, measuring output quality across retrieval and generation
- Converted PyTorch models to ONNX to speed up inference in production
- Tracked experiments and versioned models with MLflow so results were reproducible
- Deployed ML services on AWS (Lambda, ECR, S3, DynamoDB)

KAN Innovations, ML Engineer Intern — Mumbai, Maharashtra — Feb 2025 – May 2025 — 3 months

- Built a PyTorch ulcer detection model reaching about 85% accuracy
- Improved the performance of an eye posture detection model

Projects

Financial AI Agent (Demo)

Prototype agent for financial data retrieval and reasoning in Python.

- Built modular retrieval and reasoning pipelines for financial insights
- Implemented query routing and response refinement in Python

Aethelgard

Real-time gateway that analyzes LLM outputs for hallucinations and risk.

- Built a real-time analyzer using a secondary model for semantic risk scoring of LLM responses
- Designed automated hallucination detection logic and streamed risk metrics to Datadog

Achievements

Qualified the Graduate Aptitude Test in Engineering (GATE)